Data Dictionary — Nova Scotia GHG Emissions Dataset

Dataset Overview

This dataset contains cleaned greenhouse gas (GHG) emissions data for Nova Scotia, derived from Government of Canada open data sources. It has been processed for analytical use in exploratory data analysis (EDA) and Power BI dashboard development.

The dataset is structured to support:
- Time-series analysis of emissions trends
- Sector-level emissions comparison
- KPI calculations and forecasting

Table Structure

Column Name	Data Type	Description	Example
Year	int64	Reporting year for emissions data	2020
Emissions	float64	GHG emissions in kilotonnes of CO_2 equivalent	1450.75
Sectors	object	Economic sector responsible for emissions	Electric power generation

Column Name	Data Type	Description
Year	Integer (int64)	Reporting year for the greenhouse gas emissions record. Used for time-series trend analysis and forecasting.
Nova Scotia	Text (object)	Name of one of the Canadian province or territory associated with the emissions data.
Emissions	Decimal (float64)	Greenhouse gas emissions

		value measured in kilotonnes of carbon dioxide equivalent (kt CO_2e). Used for KPI calculations, trend analysis, and Power BI visualizations.
Sectors	Text (object)	Economic or industrial sector responsible for the reported emissions (e.g., transportation, agriculture, electricity generation, manufacturing). Used for sector-level emissions analysis and breakdowns.

Item	Details
Dataset Name	Canadian Greenhouse Gas Emissions Dataset
Source	Government of Canada Open Data / Statistics Canada
Time Period	2013 – 2023
Geographic Coverage	Canadian Provinces and Territories
Primary Use	Climate trend analysis, emissions comparison, Power BI dashboarding, and forecasting
Unit of Measurement	Kilotonnes of carbon dioxide equivalent (kt CO_2e)

Data Notes & Definitions

Emissions Unit (kt CO_2e): Standardized measure combining all greenhouse gases.
Sector Classification: Industry categories used for emissions breakdown.
Time Coverage: Multi-year dataset for trend analysis.

Data Cleaning Summary

- Removed unnecessary metadata columns
- Filtered to Nova Scotia only
- Standardized column names
- Converted data types
- Removed duplicates and nulls
- Excluded balancing categories
- Simplified sector labels

Analytical Use Cases

- Trend analysis
- Sector contribution analysis
- KPI calculations
- Power BI dashboard visuals
- Predictive modeling

Limitations

- Aggregated sector-level data
- Limited granularity
- External influencing factors not included

Summary

The dataset is structured, consistent, and optimized for analytical workflows, supporting data-driven environmental insights.